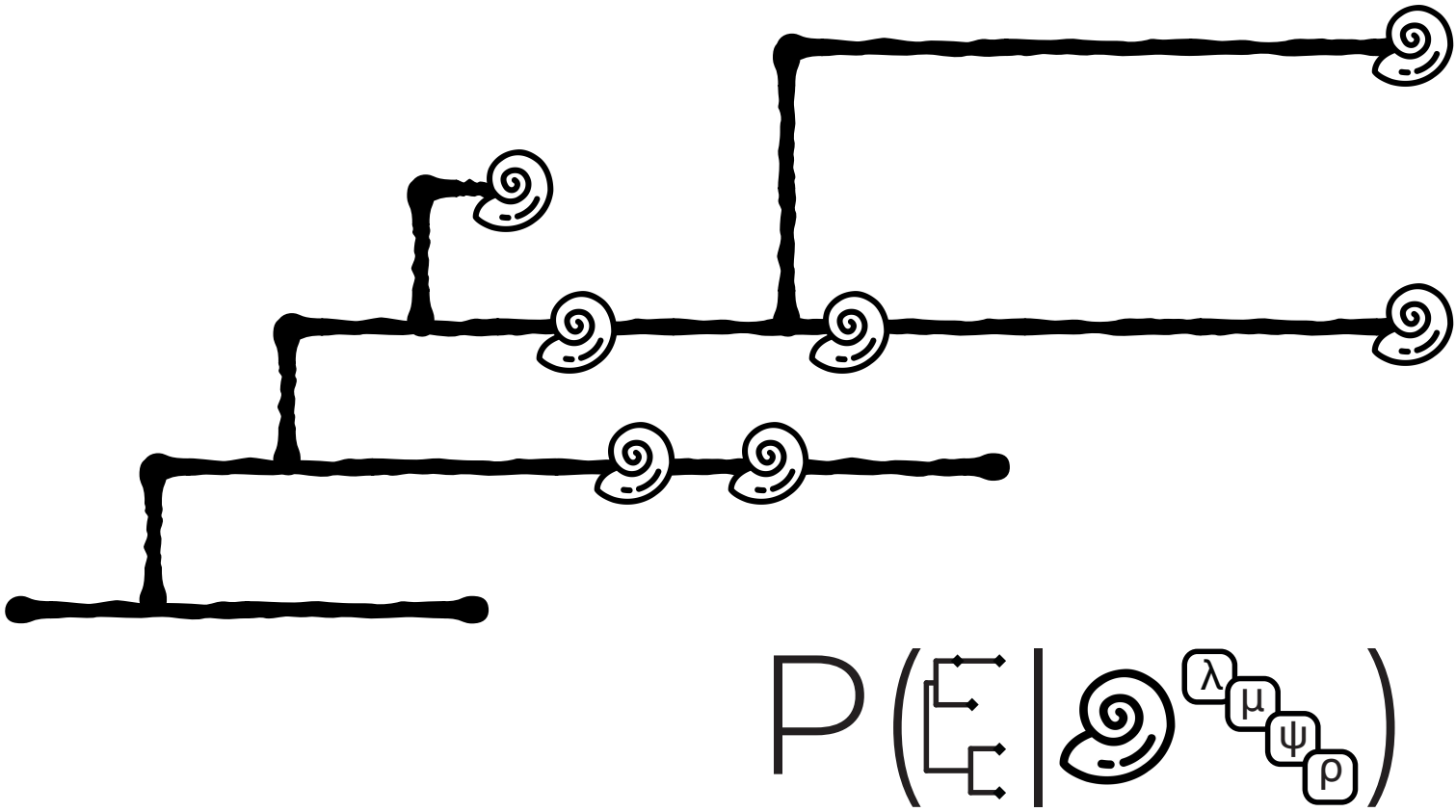
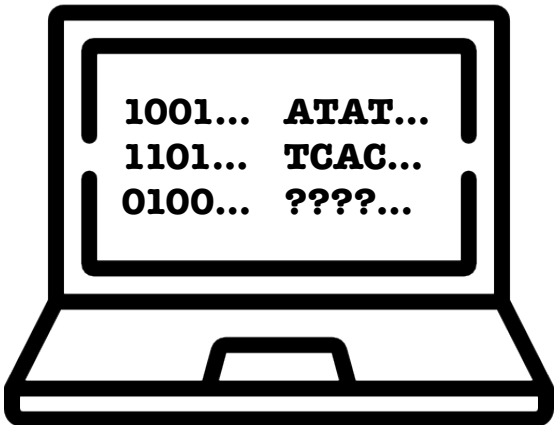
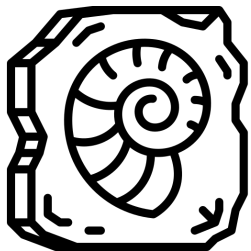


BDMM-Prime for Macroevolution

Taming the BEAST Beijing

Rachel Warnock

17.07.2025





Tim Vaughan

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Author: Tim Vaughan
Created: 2025-07-14 Mon 17:01



Source code: <https://github.com/tgvaughan/BDMM-Prime>

1. Introduction

BDMM-Prime is a BEAST 2 package for performing phylodynamic inference under a variety of linear birth-death-sampling models with/without types.

1.1. Comparison with BDMM

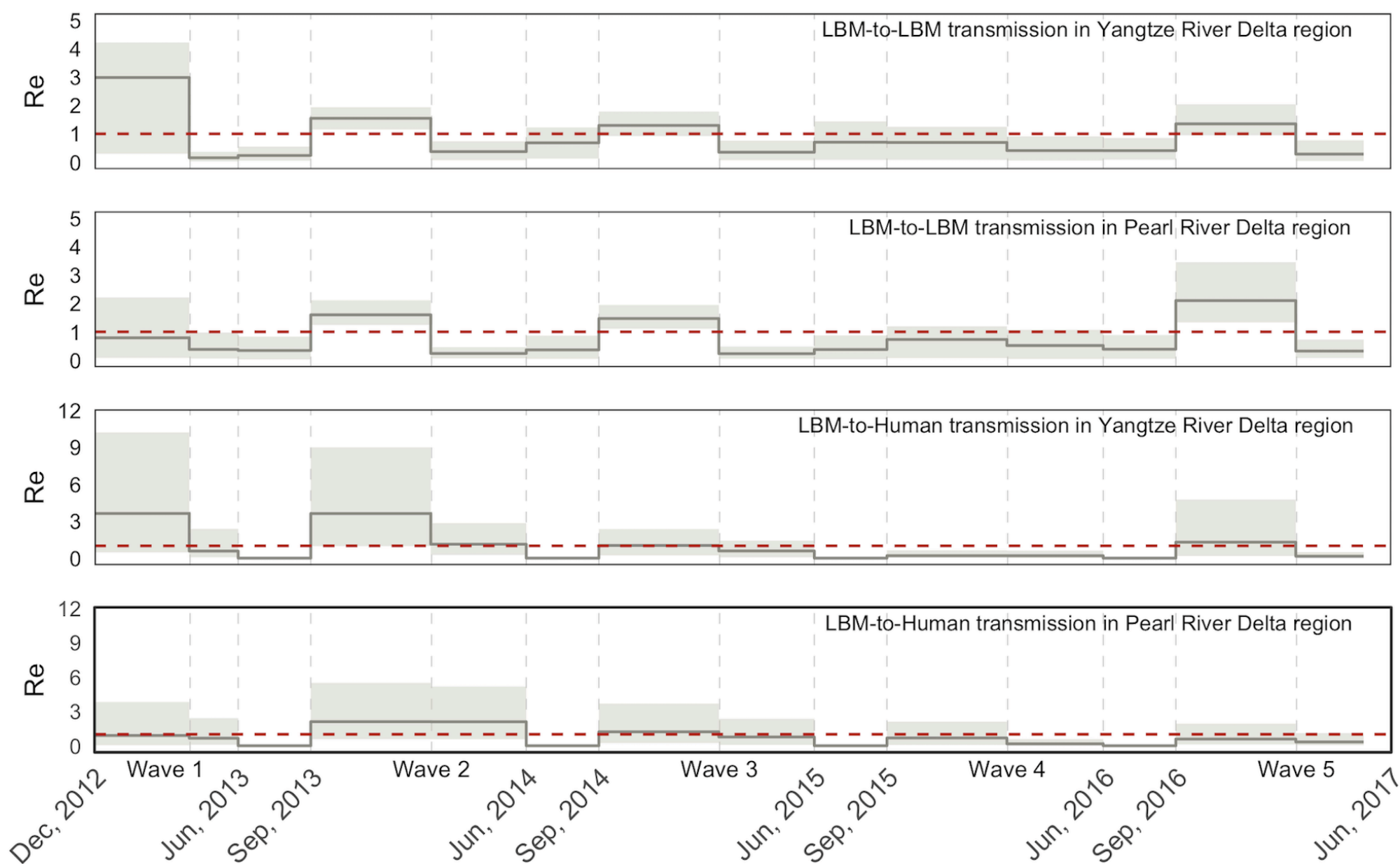
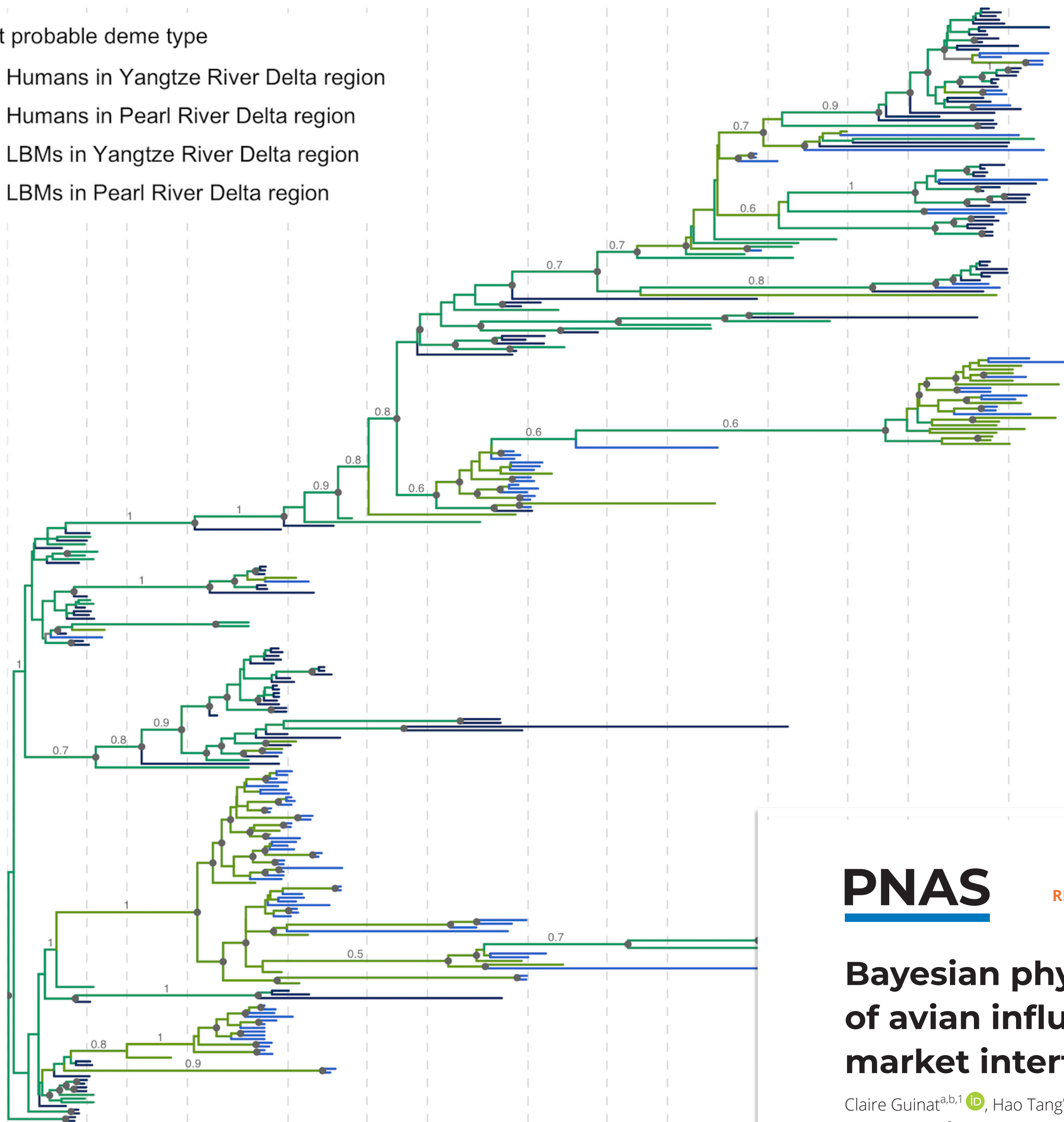
The BDMM-Prime project is a hard fork of the original [BDMM project](#). The intention is to extend the functionality of the original package, while improving its flexibility and ease of use. It incorporates the following enhancements:

- an improved BEAUti interface that allows a much more diverse range of analyses to be configured,
- automatic fall-back to analytical solutions for unstructured (single type) analyses, meaning BDMM-Prime can in principle be used in place of [BDSKY](#),
- use of stochastic mapping for efficient sampling of ancestral types

<https://tgvaughan.github.io>

Most probable deme type

- Humans in Yangtze River Delta region
- Humans in Pearl River Delta region
- LBMs in Yangtze River Delta region
- LBMs in Pearl River Delta region

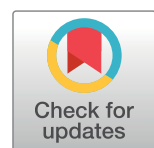


PNAS

RESEARCH ARTICLE

EVOLUTION

OPEN ACCESS

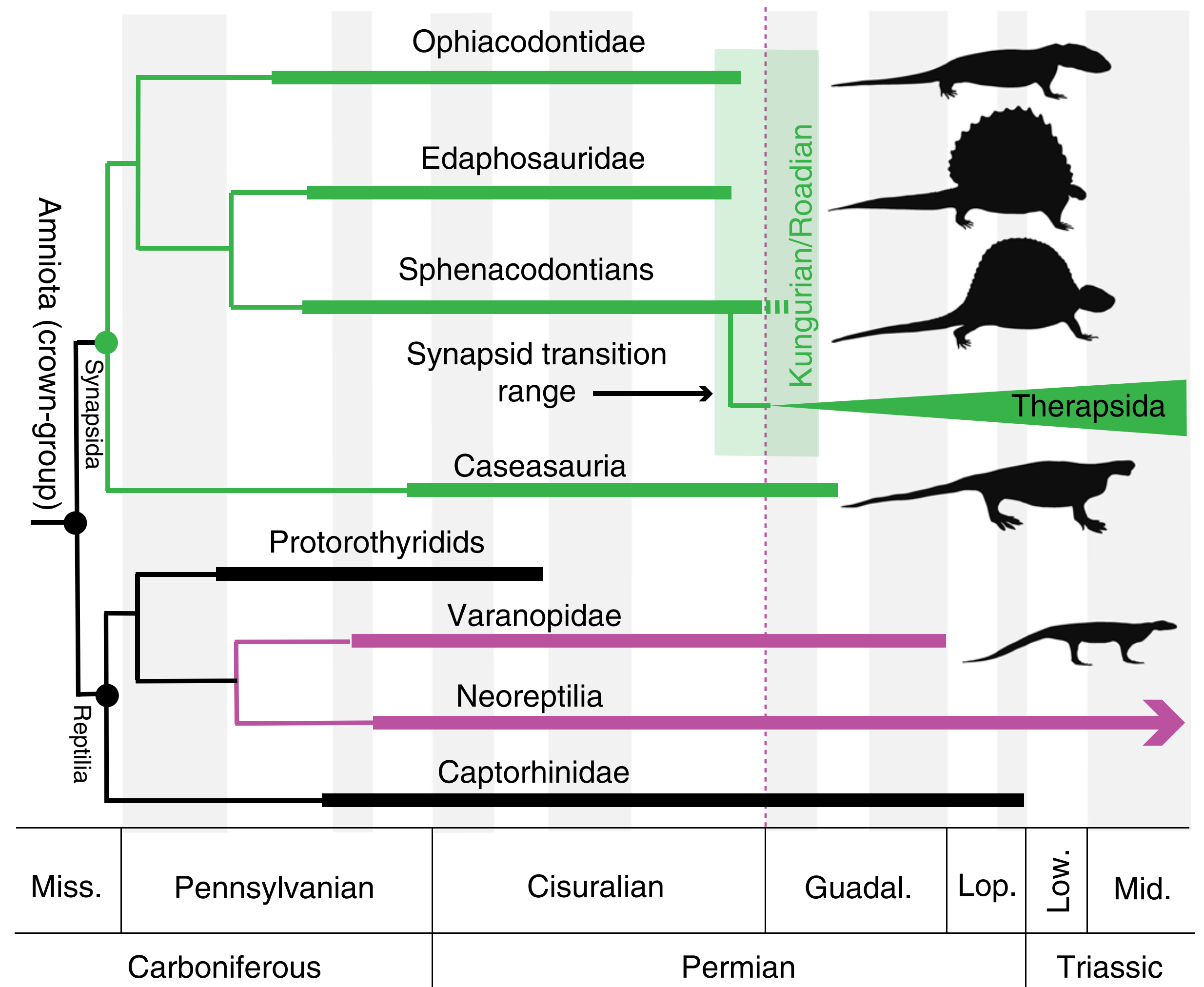


Bayesian phylodynamics reveals the transmission dynamics of avian influenza A(H7N9) virus at the human–live bird market interface in China

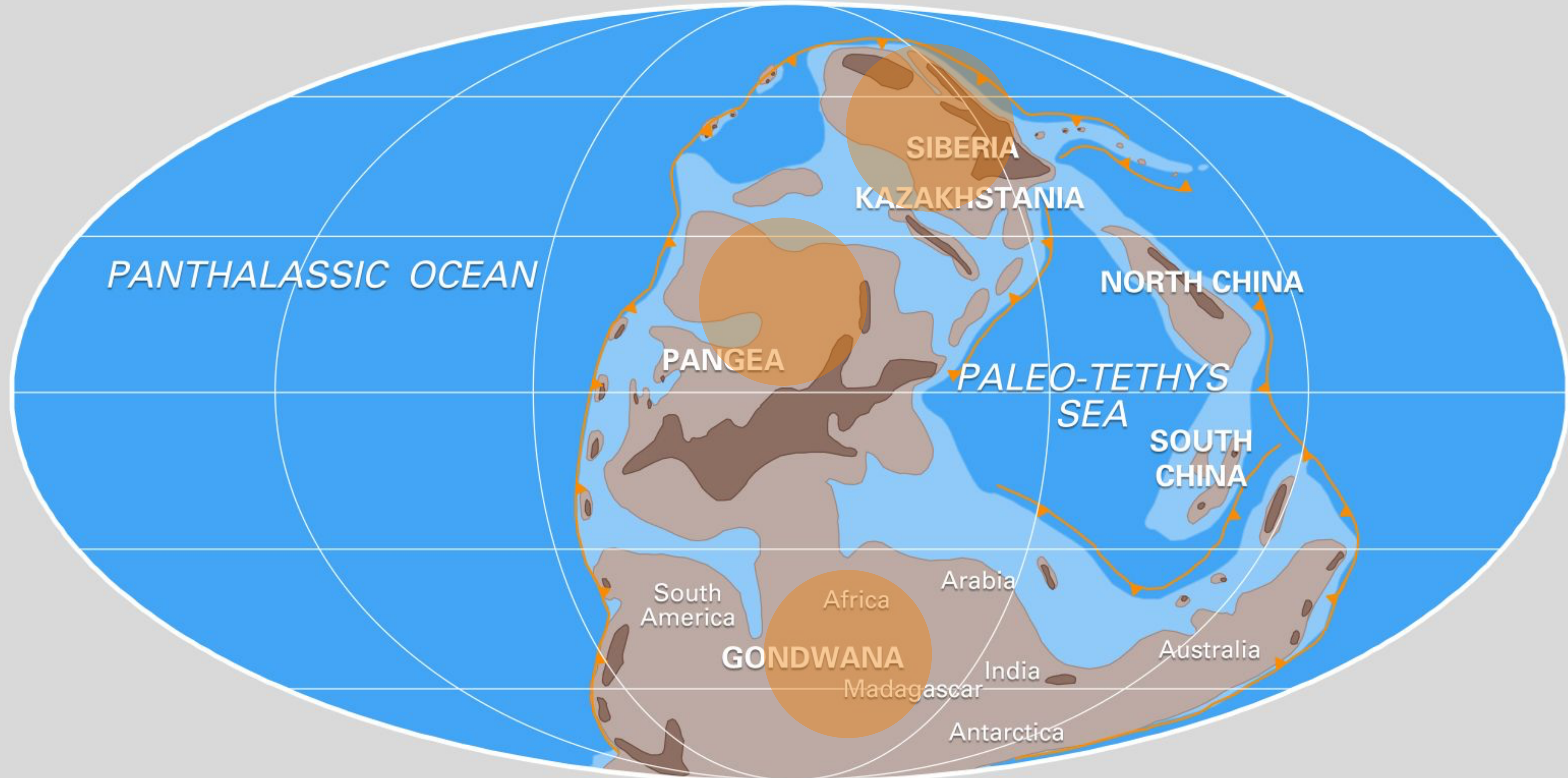
Claire Guinat^{a,b,1} , Hao Tang^c , Qiqi Yang^d , Cecilia Valenzuela Agüir^{a,b} , Timothy G. Vaughan^{a,b} , Jérémie Scire^{a,b} , Hongjie Yu^e, Wei Wang^e, Zhiyuan Chen^e, Mariette F. Ducatez^f , and Tanja Stadler^{a,b}

Exercise

Both skyline and multi-type extensions in available via BDMM-Prime



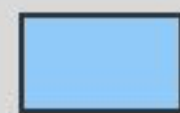
Late Carboniferous 306 million years ago



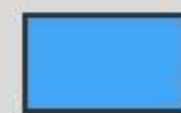
Mountains



Land



Shallow seas



Deep ocean basins

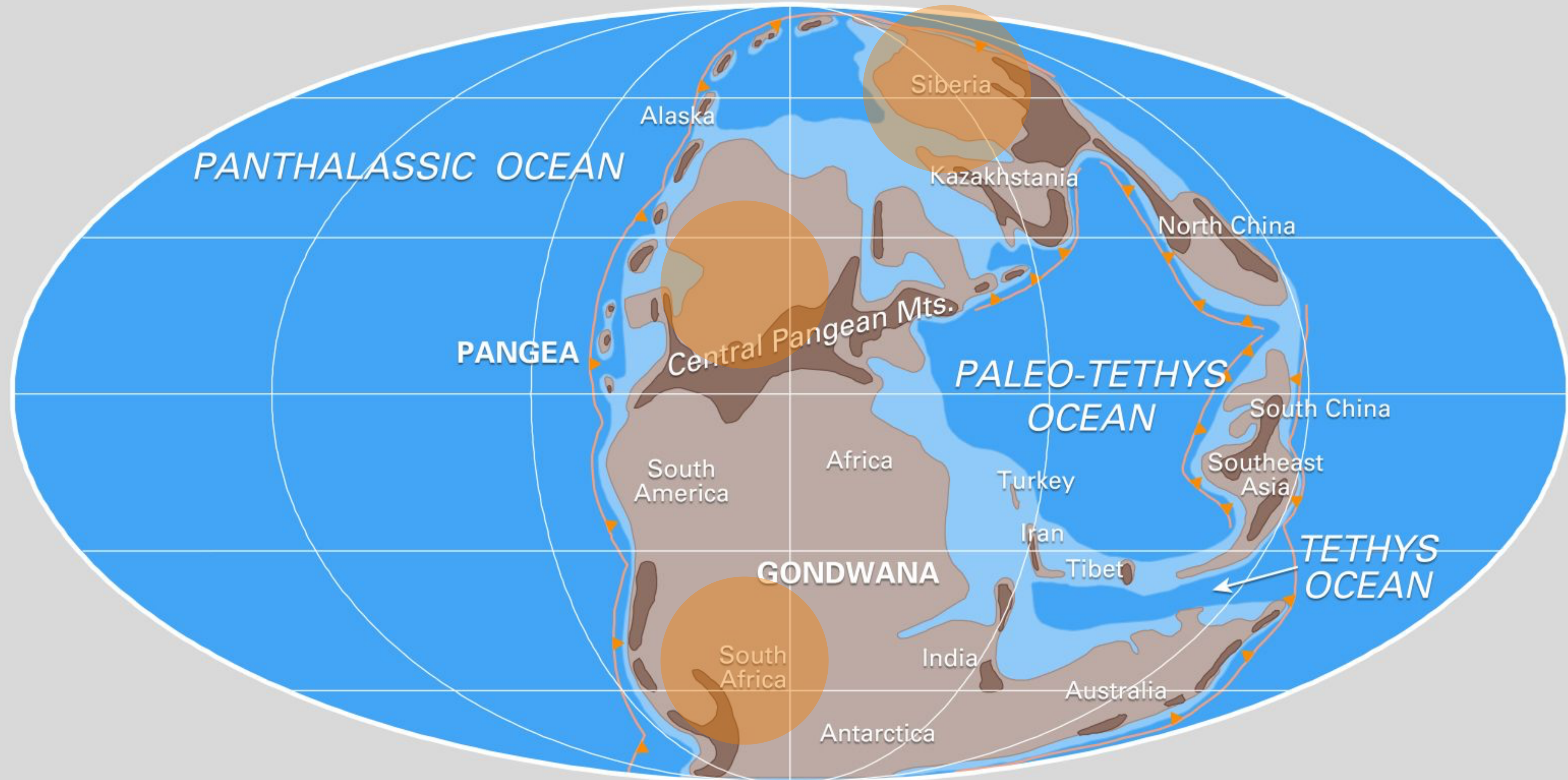


Subduction zone
(triangles point in the
direction of subduction)

SOURCE: © 2001 C.R. Scotese, PALEOMAP Project

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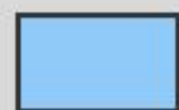
Late Permian 255 million years ago



Mountains



Land



Shallow seas



Deep ocean basins

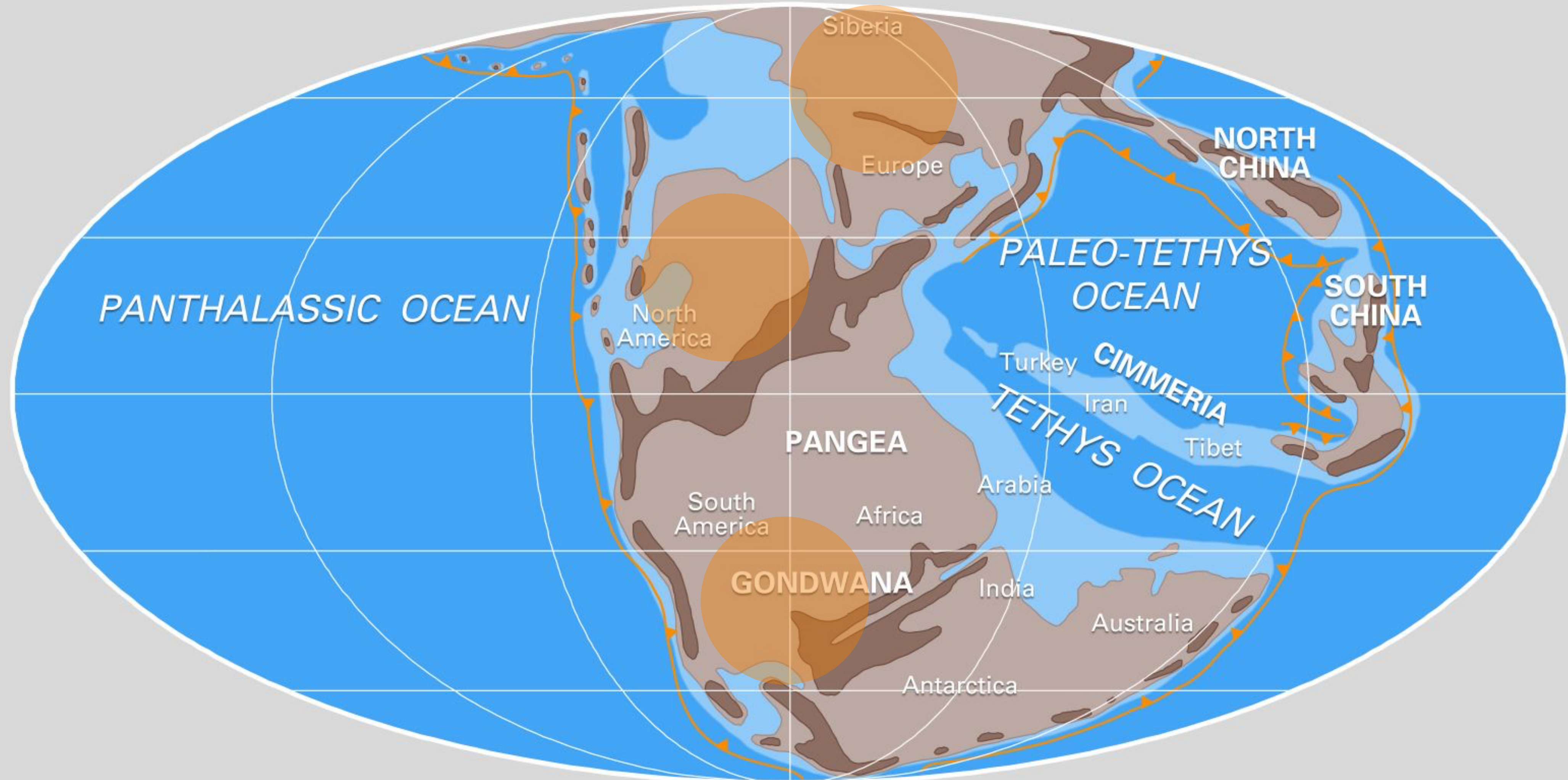


Subduction zone
(triangles point in the
direction of subduction)

SOURCE: © 2001 C.R. Scotese, PALEOMAP Project

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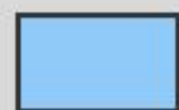
Early Triassic 237 million years ago



Mountains



Land



Shallow seas



Deep ocean basins



Subduction zone
(triangles point in the
direction of subduction)

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BDMM-Prime checklist for macroevolution

We recommend always double checking your model parameters

Parameter	Notes
Birth (speciation) Rate — λ	usually estimated (time or lineage variable)
Death (speciation) Rate — μ	usually estimated (time or lineage variable)
Sampling (fossilisation) Rate — ψ	usually estimated (time or lineage variable)
Rho (extant sampling) Probability — ρ	usually fixed
Removal Probability — r	not applicable in macroevolution, fix to zero
Migration Rate — m	usually estimated (time or lineage variable)
Birth Rate Among Demes	not applicable in macroevolution, fix to zero